LAB 1

Objective: To study and simulate the behavior of an analog computer using differential equation. (Any five)

Theory

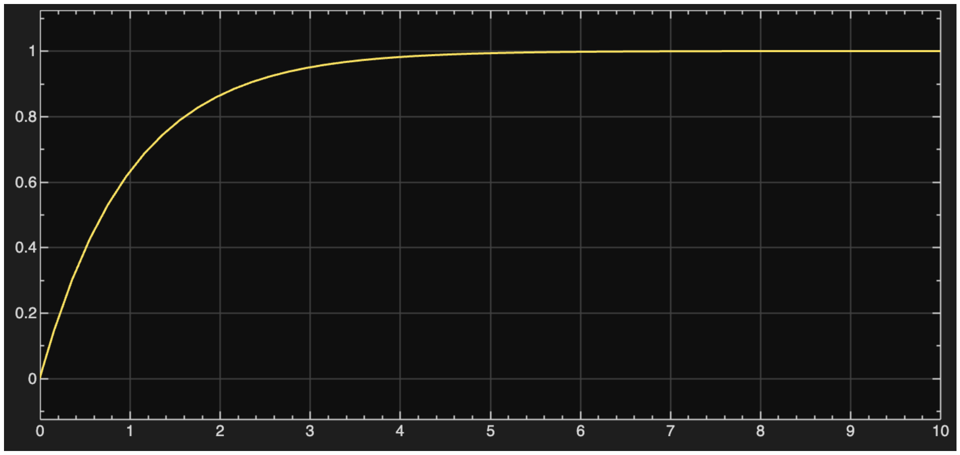
The simulation of an analog computer is based on three main concepts:

- Mathematical Modeling: Any physical system can be represented by a differential equation. In an analog computer, voltage is used as the "analog" for the variables in those equations.
- Operational Blocks: The system uses three primary operations:
  - Integration: Using Op-Amp integrators to convert derivatives (like acceleration) into variables (like velocity and displacement).
  - Summing: Combining different terms of the equation.
  - Scaling (Gain): Multiplying variables by constants (the coefficients in your equations).
- Feedback Loops: By connecting the output of the integrators back to the input summer, the circuit "forces" itself to follow the mathematical rules defined by the differential equation.

Procedure

1. Rearrange the given differential equation to isolate the highest derivative term.
2. Determine the number of Integrators required based on the order of the equation (1 for first-order, 2 for second-order).
3. Set up the Summing Junction to add the input signals (forcing functions like *sin(t)* or constants) and the feedback signals.
4. Assign Gains (coefficients) to the feedback loops using Potentiometers or Gain blocks.
5. Run the simulation and observe the behavior of *y(t)* on an oscilloscope or Time-Scope.

Experiment:



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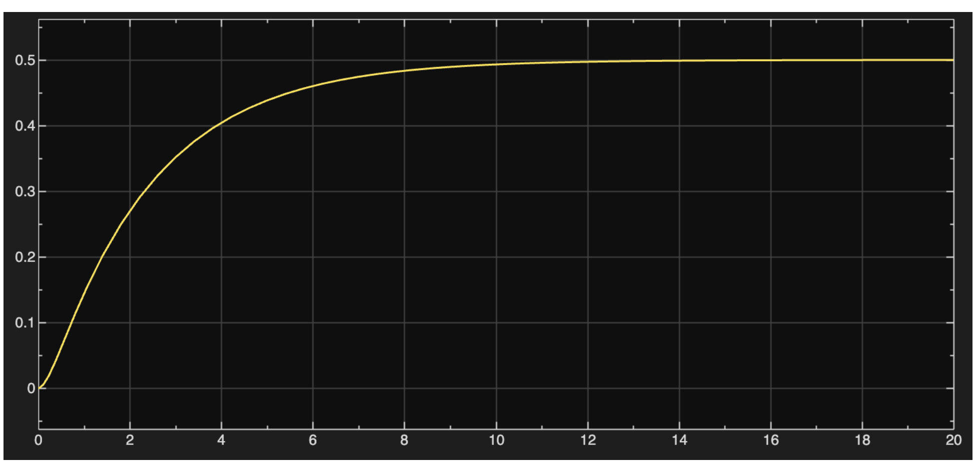

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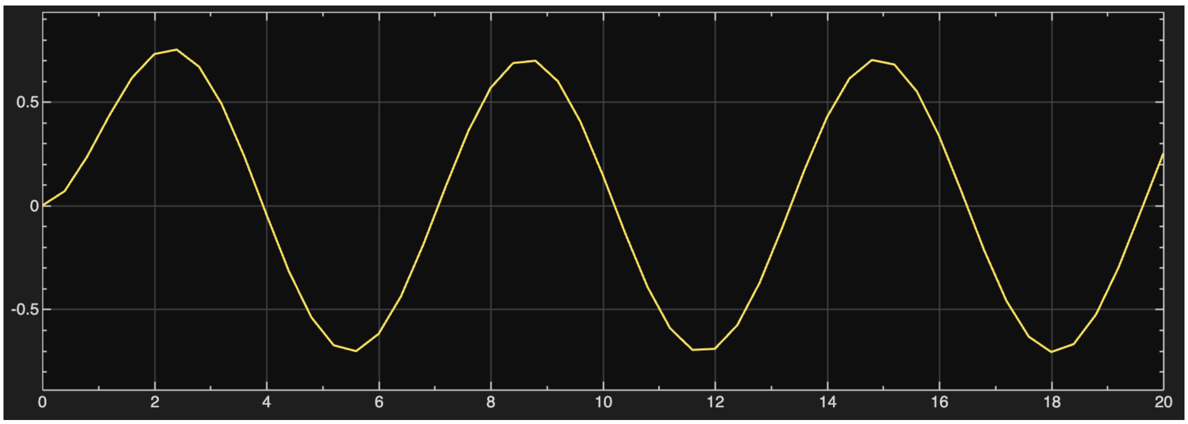
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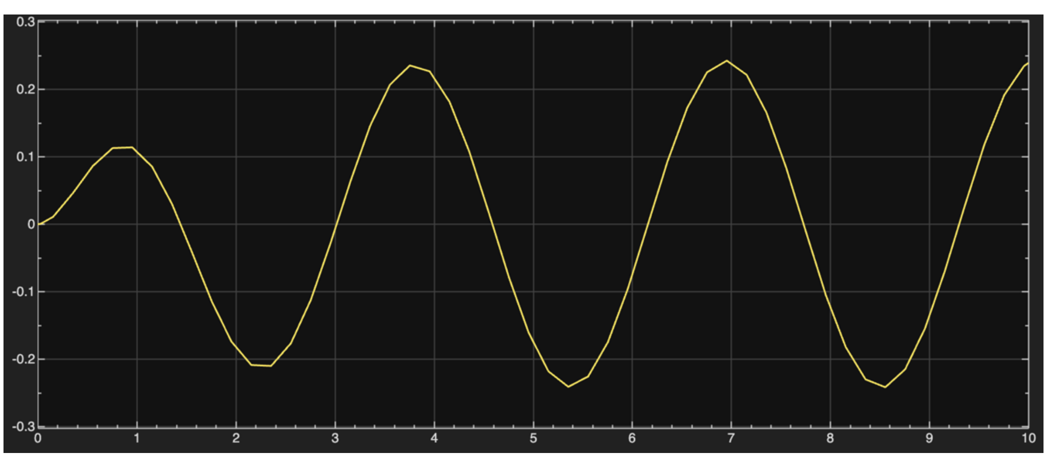
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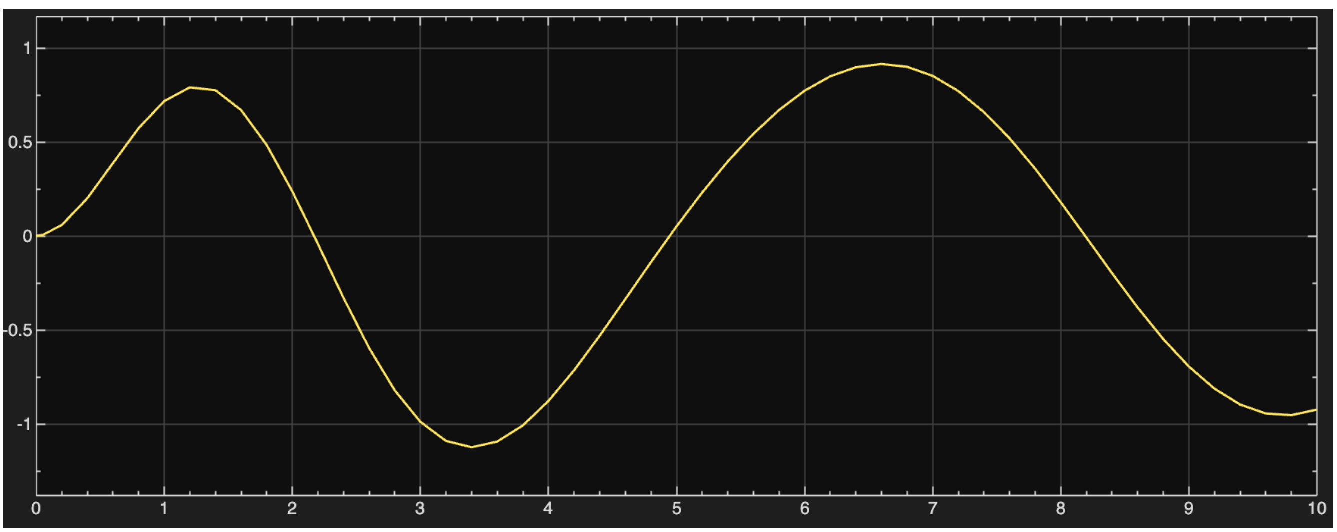

Result

The differential equations were successfully modeled. For 1st order systems, *y(t)* followed an exponential path. For 2nd order systems, the presence of the *dy/dt* term acted as "damping," while the forcing functions (*sin, cos*) dictated the long-term behavior.

Conclusion

Analog simulation provides a continuous-time solution to differential equations. By using integrators and feedback loops, complex physical systems (represented by these equations) can be simulated in real-time.